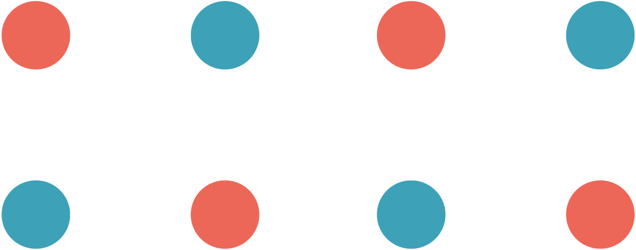
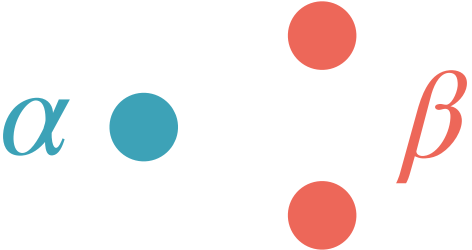
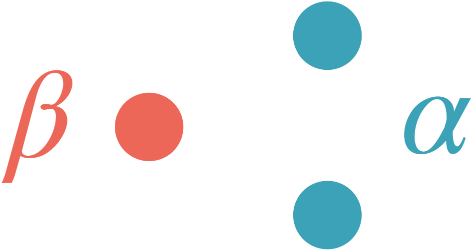


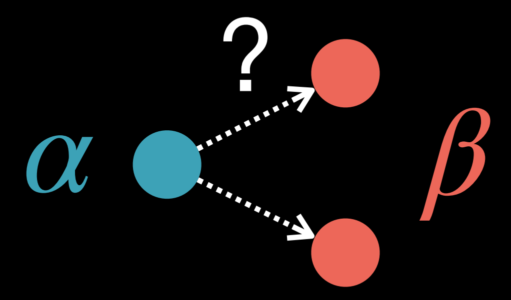

α
β


β
α

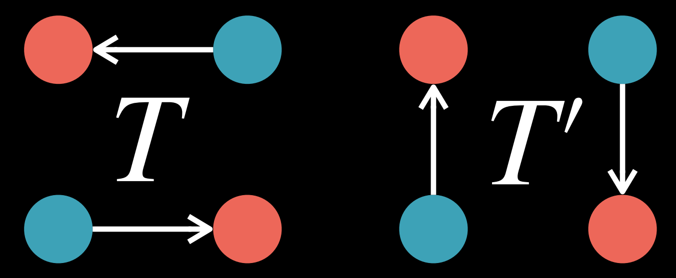
A large part of OT Theory: proving when solutions exist, are unique, and are nice

Properties of Monge Problem

- Does the solution (the Monge map) always exist?
  - No! Example:

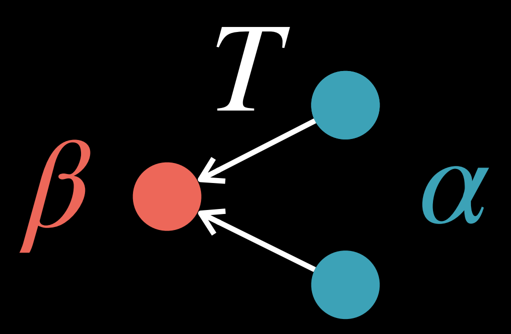


- Is the solution unique?
  - No! Example:



$\mathrm{cost}(T) = \mathrm{cost}(T')$

- Is the problem symmetric on α and β?
  - No!



- Is the optimization problem convex?
  - No! Highly non-convex, set of push forwards is non-convex
  - Makes numerical solution fragile, especially in high dim
- Suppose the solution exists. Is it “nice”?
  - No! It may be e.g. discontinuous
  - In some settings (e.g., convex cost + regular densities), the solution is smooth

A large part of OT Theory: proving when solutions exist, are unique, and are nice

Brenier's Theorem

Assumptions:

1. $c(x, y) = \|x - y\|^2$

2. α has a density $\rho_\alpha = \frac{\mathrm{d}\alpha}{\mathrm{d}x}$

Monge's Problem:

(Continuous / general version)

$$\min_{\beta = T_\sharp \alpha} \int_X c(x, T(x)) \mathrm{d}\alpha(x)$$